

# YELDEN PROTOCOL

## Accountability for Autonomous Trading Agents

*“An agent that cannot lose cannot be trusted.” — Taleb, Skin in the Game*

Version 16.0 · October 2026 · Polygon Mainnet (Chain ID 137)

AIAgentRegistry

**0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E**

verify: [polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E](https://polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E)

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## 00 — Executive Summary

Autonomous trading agents manage capital at scale. None are accountable.

An agent that underperforms today can simply stop, restart under a new address, and continue operating with no track record, no consequence, and no mechanism for capital allocators to distinguish skill from luck. This is not a theoretical problem — it is the current state of the entire agent economy.

Yelden Protocol is the accountability layer for autonomous trading agents. We provide:

- YAAF — a multi-component, literature-grounded score validated empirically to predict future returns.
- AI-Agent-Registry — an on-chain registry on Polygon that stores scores, enforces fees, and slashes underperformers.
- A fee model in USDC/USDT/MATIC where excellence is free and mediocrity costs.

The core empirical result: YAAF's S\_RAW score demonstrated an Information Coefficient of  $\rho=0.4362$  ( $p=0.0005$ ,  $n=60$ ) against future returns on GMX V2 with real R-multiples. Sharpe ratio on the same sample achieved only  $\rho=0.1747$  ( $p=0.18$ ). YAAF is 2.5x more predictive than Sharpe.

Current state: 4 live trading agents in production (Pepperstone, FTMO, OKX, IBKR), 1,007 addresses examined by the observatory across two venues and 425 of them scored, AI-Agent-Registry deployed on Polygon mainnet.

## 01 — The Problem

Three structural failures define the current agent economy.

### 1.1 Agents manage capital. None are accountable.

Autonomous trading agents execute real capital across DeFi, FX, and equities. They are faster than humans, run continuously, and operate without oversight. But there is no infrastructure that holds them accountable when they fail.

### 1.2 Track records do not survive.

An agent that loses money can terminate and restart under a new address. Its history is erased. No allocator can verify performance over time. There is no persistent, verifiable, cross-venue track record.

### 1.3 No common scoring standard.

Every protocol has its own metrics. Sharpe here, ROI there, win rate elsewhere. There is no canonical score that is grounded in the quantitative finance literature, scale-free, anti-overfit, anti-gaming, and empirically validated against future returns.

## 02 — What Yelden Solves

Yelden introduces an accountability-first framework where every economic flow is conditioned on verifiable performance.

| Problem                    | Yelden's solution                                     |
|----------------------------|-------------------------------------------------------|
| Agents without consequence | slash() on verified drawdown / capital loss           |
| No persistent track record | SISTEMA score compounds over rounds via EMA           |
| No common standard         | YAAF — 11 components, literature-grounded             |
| No anti-gaming             | Volume Axiom — hard gate on real position size        |
| No empirical validation    | IC test: $\rho=0.4362$ , $p=0.0005$ (GMX V2, $n=60$ ) |

## 03 — The YAAF Score

### 3.1 S\_RAW v5 — 11 Components + 2 Penalties

S\_RAW is computed from 11 normalised performance metrics. The weights below are the calibrated values of the Formal Specification; they sum to 0.96 and are divided by that sum before being applied, so that S\_RAW is bounded in [0, 100] while relative importance is preserved exactly (Amendment v5.0.1). CVaR and Drawdown enter as direct subtractions from the weighted sum.

| #  | Component               | Weight | Formula                                                                  |
|----|-------------------------|--------|--------------------------------------------------------------------------|
| 1  | Sharpe × DSR            | 0.14   | $\min(\text{SR}/2.5, 1) \times 100 \times \text{DSR}$                    |
| 2  | Sortino × CF            | 0.07   | $\min(\text{Sortino}/8, 1) \times 100 \times \min(\text{n\_loss}/20, 1)$ |
| 3  | Win Rate                | 0.08   | $\text{WR} \times 100$                                                   |
| 4  | Profit Factor           | 0.10   | $\min((\text{PF}-1)/1.5, 1) \times 100$                                  |
| 5  | Avg R                   | 0.07   | $\min(\text{AvgR}/1.5, 1) \times 100$                                    |
| 6  | Expectancy              | 0.09   | $\min(\max(\text{E}(\text{R}), 0)/0.6, 1) \times 100$                    |
| 7  | Inv. Volatility         | 0.05   | $\max(1 - \sigma_d/2, 0) \times 100$                                     |
| 8  | Stability <sup>2</sup>  | 0.09   | $(1/(1+\sigma_{\text{mSharpe}}))^2 \times 100$                           |
| 9  | Smoothness <sup>2</sup> | 0.08   | $\min(\text{Calmar}/4, 1)^2 \times 100$                                  |
| 10 | Pf Percentile           | 0.12   | % of 1,000 random portfolios beaten                                      |
| 11 | Mc (Independence)       | 0.07   | $(1 -  \beta )^2 \times 100$                                             |
| —  | CVaR penalty            | —      | $\min(\text{CVaR}/2, 1) \times 15$                                       |
| —  | Max DD penalty          | —      | $\text{MaxDD}\% \times 0.20$                                             |

#### Component Specifications

Four of the components above admit more than one reading. The Formal Specification fixes them as follows, so that any third party holding the same trade record can reproduce a published score.

**Pf Percentile (c10).**  $K = 1,000$  vectors are drawn from  $N(0, \sigma_r)$  with the same length as the agent's R-multiple series. The component is the percentage of those random portfolios whose annualised Sharpe the agent's real Sharpe exceeds.  $\text{Pf}\% = 97$  means the strategy beats 97% of random portfolios with identical risk and sample size.

**Mc — Independence (c11).**  $\beta$  is the correlation between the agent's daily aggregated R and the daily returns of the underlying market. The quadratic form rewards edge that is independent of market direction, and penalises a track record that is mostly market beta.

**CVaR penalty.** CVaR at  $\alpha = 0.95$  is the expected shortfall of the R-multiple series on the loss side, taken positive. Coherent and tail-sensitive, it is preferred over VaR for fat-tailed trading returns.

**SF — Safety Factor.** The “last 8 S\_RAW” are the eight most recent scoring rounds, carried in the agent's persisted state — not eight slices of a single submission. Each round contributes one EMA update, from a cold-start prior of  $\text{EMA}_0 = 300$ .

#### Design Principles

- Scale-free — all metrics in R-multiples or dimensionless ratios.
- Anti-overfit — DSR (Bailey & López de Prado, 2014) penalises statistical noise.
- Tail-aware — CVaR (Rockafellar & Uryasev, 2000) as explicit penalty.
- Anti-gaming — Volume Axiom is a hard gate: failure forces S\_RAW = 0.

### 3.2 SISTEMA — $\text{EMA} \times \text{CF} \times \text{SF}$

SISTEMA is the time-compounding score that governs all protocol economics, bounded in  $[0, 1000]$ .

$$\text{SISTEMA}(t) = \text{EMA}(t) \times \text{CF}(t) \times \text{SF}(t)$$

| Factor | Formula                                                                              | Meaning                                |
|--------|--------------------------------------------------------------------------------------|----------------------------------------|
| EMA    | $\text{EMA}(t) = 0.85 \times \text{EMA}(t-1) + 0.15 \times S_{\text{RAW}} \times 10$ | Memory of past performance             |
| CF     | $\min(\sqrt{(\text{trades}/250)}, 1)$                                                | Confidence factor — full at 250 trades |
| SF     | $\min(\text{mean}(\text{last } 8 S_{\text{RAW}})/45, 1) \times (1 - \sigma_s/25)$    | Safety — penalises inconsistency       |

#### Score Bands

| Band         | SISTEMA  | Interpretation            |
|--------------|----------|---------------------------|
| EXPERIMENTAL | 0–199    | Insufficient track record |
| PROMISING    | 200–399  | Emerging performance      |
| VERIFIED     | 400–599  | Consistent performer      |
| ELITE        | 600–799  | Top-tier                  |
| LEGENDARY    | 800–1000 | Exceptional               |

**Reproducibility.** Every random stream used by the Pf Percentile is seeded deterministically, so an identical trade record always yields an identical score. A score committed on-chain can therefore be recomputed and verified by any party holding the same trades — the score is auditable, not merely asserted.

### 3.3 Empirical Validation

The YAAF score was tested against future returns across three venues with different data qualities.

| Venue       | Data Quality           | n   | IC (YAAF)         | IC (Sharpe)     | IC (WR)         |
|-------------|------------------------|-----|-------------------|-----------------|-----------------|
| GMX V2      | high (real collateral) | 60  | 0.4362 (p=0.0005) | 0.1747 (p=0.18) | 0.2139 (p=0.10) |
| Hyperliquid | medium (R estimated)   | 268 | -0.0305 (p=0.62)  | 0.1144 (p=0.06) | 0.0171 (p=0.78) |
| Darwinex    | low (Risk Engine)      | ~80 | 0.1411 (p<1e-30)  | —               | —               |

The methodological finding: YAAF's predictive power is directly proportional to the quality of the R-multiple input. When the collateral is explicit (GMX V2), YAAF is 2.5x more predictive than Sharpe. When estimated, the edge vanishes. This validates the Volume Axiom as a fundamental gate, not a decorative constraint.

### 3.4 Data Quality Tiers

Every score in Yelden carries an explicit data quality tier.

| Tier   | Definition                               | Multiplier |
|--------|------------------------------------------|------------|
| high   | Real collateral / R-multiple from source | 1.00       |
| medium | R-multiple estimated from notional       | 0.50       |
| low    | Returns smoothed by intermediary         | 0.25       |

```
C_allocated = min(C_kelly, C_stage) × quality_multiplier
```

## 04 — The AIAgentRegistry

The AIAgentRegistry is the on-chain enforcement layer for YAAF scores. Deployed on Polygon Mainnet.

**Registry address:** `0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E`

verify: [polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E](https://polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E)

### 4.1 registerAgent()

An agent registers by submitting its address and collateral stake in USDC. The registry stores: agent address, data source, data quality tier, registration timestamp, current SISTEMA score, and score history.

### 4.2 updateScore()

Scores are updated by addresses holding SCORER\_ROLE. Each update writes S\_RAW (current round), SISTEMA (compounded), Stage, and Timestamp.

### 4.3 slash()

The slash mechanism fires on either drawdown or capital loss, whichever produces the higher slash amount. This closes the “24% loophole” — an agent that stops trading at 24% DD to avoid a 25% threshold is still slashed on the capital loss dimension.

| Drawdown | Capital Loss | Slash (stake)                                |
|----------|--------------|----------------------------------------------|
| < 10%    | < 10%        | 0                                            |
| 10–25%   | < 10%        | $0.4 \times (DD - 0.10) \times \text{stake}$ |
| 24%      | 20%          | capital loss trigger overrides               |
| ≥ 25%    | any          | full slash                                   |

### 4.4 isEligible() — Integration Contract

Any protocol, vault, or fund can integrate Yelden in two lines:

```
IAgentRegistry registry = IAgentRegistry(  
    0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E  
);  
require(registry.isEligible(agent), "Agent not eligible");
```

isEligible() returns true when: SISTEMA ≥ 400 (VERIFIED or above), data quality tier is high or medium, and no pending slash event.

## 05 — Fee Model

Yelden operates entirely in USDC, USDT, and MATIC. No native token. No governance. No UBI.

### 5.1 Monthly Fee

$$\text{monthly\_fee} = \text{base\_rate} \times (1000 - \text{SISTEMA}) / 1000$$

At a base rate of 10 USDC/month:

- SISTEMA 1000 => 0 USDC (excellence is free)
- SISTEMA 500 => 5 USDC
- SISTEMA 0 => 10 USDC (self-eliminates)

Fees flow to the protocol treasury.

### 5.2 Slash Schedule

| Level      | Trigger            | Penalty                        |
|------------|--------------------|--------------------------------|
| WARNING    | Grade 1 misconduct | 10% of stake                   |
| SUSPENSION | Grade 2 misconduct | 50% of stake                   |
| BAN        | Critical fraud     | 100% of stake + deregistration |



## 06 — Live Validation

### 6.1 Observatory — 1,007 Addresses Examined

Yelden operates a continuous observatory across two venues. Discovery is broad and the scoring gate is strict: an address must have at least 30 closed trades before it receives a score at all.

| Venue       | Addresses examined | YAAF scored | MM scored | Below minimum | Data quality |
|-------------|--------------------|-------------|-----------|---------------|--------------|
| Hyperliquid | 562                | 231         | 134       | 197           | medium       |
| GMX V2      | 445                | 60          | —         | 385           | high         |
| Total       | 1,007              | 291         | 134       | 582           | —            |

582 of 1,007 addresses — 58% — fall below the 30-trade minimum and carry no score. On GMX V2 the proportion reaches 87%. This is the sample-size gate doing its work, not a coverage failure: a score computed on twelve trades would be noise presented as a measurement. The two venue sets are disjoint, so no address is scored twice.

Scores update every 6 hours. The observatory is the empirical foundation for YAAF parameter calibration.

### 6.2 Two Scorers, Two Profiles

The observatory runs two distinct scoring engines, because two distinct kinds of agent earn in two distinct ways.

**YAAF** scores directional traders and scalpers — agents whose edge appears as discrete R-multiples on closed positions. 291 addresses. Scale-free by construction, gated by the Volume Axiom, bands EXPERIMENTAL through LEGENDARY.

**MM Score** scores market makers and hedgers — agents that hold permanent inventory and earn from rebate, spread and funding rather than from discrete closed trades, which the R-multiple model cannot represent. 134 addresses. Nine components (Sharpe, drawdown, maker share, rebate per volume, realised per volume, funding, equity size, market diversification and fill activity) with weights summing to 1.00, and a hard gate at equity below \$1,000 with fewer than 100 fills. Bands INACTIVE through PRIME.

**The two scores are not comparable.** Both report on 0–1000, but YAAF is scale-free while the MM Score carries an explicit size term —  $\log_{10}$  of equity, capped at \$100M — so a larger market maker scores higher partly for being larger. The MM Score also applies neither the Deflated Sharpe Ratio nor the CVaR penalty. An address is scored by one engine or the other, never both, and a reader should not rank an MM Score against a SISTEMA.

### 6.3 Live Trading Agents

Four autonomous agents run in production:

| Agent           | Venue               | Status                                       |
|-----------------|---------------------|----------------------------------------------|
| Pepperstone bot | cTrader             | Live — 16 closed trades                      |
| FTMO bot        | cTrader             | Live — 9 closed trades                       |
| OKX balancer    | OKX                 | Live — rebalancing BTC/ETH/SOL/AVAX/gold/USD |
| IBKR equities   | Interactive Brokers | Live — US stocks                             |

All four will be registered in the AIAgentRegistry as they accumulate  $\geq 30$  closed trades, at which point YAAF produces statistically meaningful scores.

## 6.4 The First Validation — Markowitz Bot

The first agent to run under YAAF was the Markowitz Bot — a multi-strategy system executing across 7 sub-strategies with Mean-Variance Optimization. It logged 122 live trades with Sharpe 1.44 and max drawdown 6.81%. Its SISTEMA trajectory demonstrated that a real agent could be scored, tracked, and compared against alternatives. The Markowitz Bot has since been superseded by the cTrader agents above, which use the same scoring pipeline.

## 07 — Data Sources & Quality

Yelden integrates with venues that expose the data YAAF needs. The critical requirement is a real R-multiple — a PnL measured against the actual capital at risk.

| Source        | R-multiple basis                    | Quality |
|---------------|-------------------------------------|---------|
| GMX V2        | initialCollateralDeltaAmount (real) | high    |
| cTrader API   | Stop-loss and MAE (real)            | high    |
| Hyperliquid   | Notional $\times$ 1% (estimated)    | medium  |
| dYdX          | Notional $\times$ 1% (estimated)    | medium  |
| Darwinex Zero | Equity curve (smoothed)             | low     |

## 08 — Roadmap

| Phase                  | Timeline | Milestone                                                                                                                                                       |
|------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foundation             | Complete | YAAF v5 · AI-AgentRegistry on Polygon · Observatory (1,007 addresses, 425 scored) · <a href="https://github.com/yeldenfund/yaaf">github.com/yeldenfund/yaaf</a> |
| cTrader Integration    | Q4 2026  | cTrader API collector · Pepperstone + FTMO registered on-chain                                                                                                  |
| Multi-Agent Onboarding | Q1 2027  | 10+ agents registered · first external agents                                                                                                                   |
| Genesis Scorer Program | Q1 2027  | 3–5 independent scorers · multi-scorer consensus                                                                                                                |
| Formal Audit           | Q2 2027  | Code4rena or Sherlock audit                                                                                                                                     |
| Chainlink Oracle       | Q2 2027  | Decentralised score submission                                                                                                                                  |

## 09 — Known Limitations

We document known limitations honestly. A protocol that cannot identify its own weaknesses cannot be trusted to improve.

- Single scorer. Currently the founder holds SCORER\_ROLE. The Genesis Scorer Program is the fix (Q1 2027).
- No formal audit. Code is pre-audit. Code4rena / Sherlock engagement planned for Q2 2027.
- cTrader integration in progress. The Pepperstone and FTMO agents are live but not yet scored on-chain.
- Data quality variance. Scores from Hyperliquid and Darwinex carry explicit quality penalties.
- Short track record. The current agent fleet has 25 closed trades across two cTrader accounts.
- cTrader R-multiple via MAE. Maximum Adverse Excursion is used as a proxy for risk when stop-loss data is not explicitly available.

## 10 — Team

### Paulo Longen — Founder

For 30 years, Paulo commanded systems where a wrong decision had real consequences — people, aircraft, missions. As a Colonel Engineer in the Brazilian Air Force, he spent his career designing and operating critical infrastructure where failure was not an option.

He retired in 2024 and entered the world of algorithmic trading. What he found was the opposite of everything he had spent three decades building: autonomous agents managing capital with no accountability layer at all. No skin in the game. No mechanism to distinguish skill from luck. No consequence for failure.

*“I spent 30 years building systems that could not fail silently. Autonomous trading agents are the opposite: they fail silently, restart under a new address, and nobody notices. Yelden closes that gap.”*

Yelden Protocol is the infrastructure he wished existed. Validated empirically ( $\rho=0.44$ ,  $p<0.001$  on GMX V2), deployed on Polygon mainnet, with four live trading agents running in production.

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## 11 — References

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## 12 — Disclaimer

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The AIAgentRegistry address cited in this document is deployed on Polygon Mainnet and has not been formally audited. Users interacting with it do so at their own risk.

**Yelden Protocol — [yelden.fund](https://yelden.fund) — October 2026**